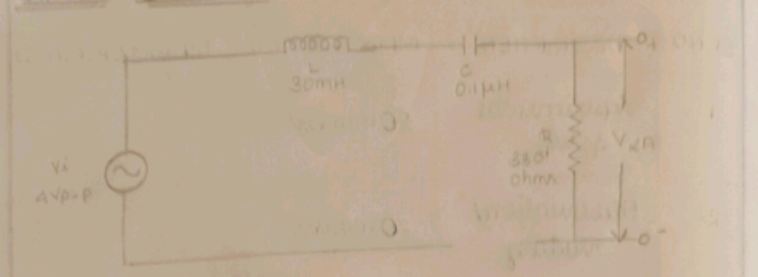
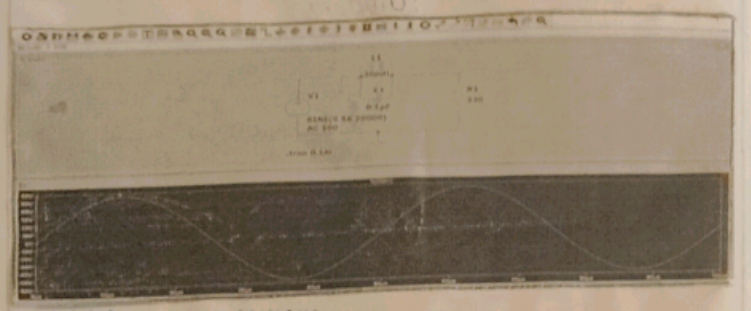
CIRCUIT DIAGRAM:



SERIES RLC CIRCUIT:



PARALLEL RLC CIRCUIT:

Ex. No. 07	FREQUENCY RESPONSE OF SERIES AND
Date 27/08/2024	PARALLEL RESONANCE CIRCUITS.

AIM:

To study the behaviour of series and parallel RLC circuits at resonance and to determine the resonant frequency, bandwidth and Q of the given RLC circuit using LTspice simulator.

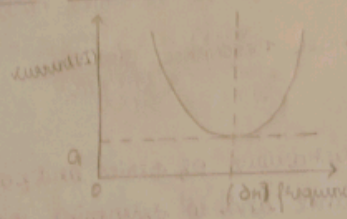
APPARATUS REQUIRED:

Sl.No.	COMPONENTS	SPECIFICATIONS	QUANTITY
1.	CRO	10 MHz	1
2.	DIB	---	1
3.	Capacitors	0.1μf	1
4.	Resistors	330Ω	1
5.	Bread board	---	---
6.	Connecting wires	---	---
7.	PC with LTspice simulator	---	---

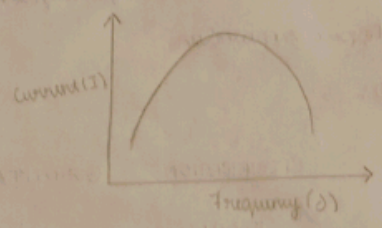
PROCEDURE:

i) Construct circuit shown in figure 1 on bread board.

PARALLEL RESONANCE:



SERIES RESONANCE:



TABULATION:

SERIES RESONANCE		PARALLEL RESONANCE	
FREQUENCY (Hz)	OUTPUT CURRENT (mA)	FREQUENCY (Hz)	OUTPUT CURRENT (mA)
500Hz	0.04mA	500 Hz	2.9mA
1kHz	0.16mA	1kHz	2.54mA
1.5KHz	0.46mA	1.5KHz	1.96mA
2KHz	1.15mA	2KHz	1.26mA
2.9kHz	8mA	2.9kHz	0.002mA
3.5kHz	2.18mA	3.5KHz	0.6mA

CALCULATION:

Theoretical calculation:

For series

$f_r = \frac{1}{\sqrt{LC}}$

ii) Connect CRO Ch1 to input and Ch2 to output

iii) Set the input voltage to 4Vp-p.

iv) Vary the frequency from 500Hz to 3KHz in small steps to get a maximum output voltage magnitude. The frequency at this maximum voltage Vm is the resonance frequency.

v) If $V_{cutoff} = V_m/\sqrt{2}$, vary the frequency again until the output voltage equals V_{cutoff}. This frequency is the cut off frequency. There should be 2 cut off frequencies on either side of resonant frequency.

vi) Calculate the bandwidth by subtracting the 2 cut-off frequencies.

vii) Calculate the Q factor.

viii) Repeat steps 1 through 7 for circuit shown in Figure 2.

FORMULAE:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$\text{Band width} = f_2 - f_1$$

$$Q = \frac{f_0}{BW}$$

$$= \frac{1}{2\times\pi\sqrt{80\times10^{-3}\times0.1\times10^{-6}}}$$

$f_r = 2.9\,kHz$

Bandwidth $= \frac{R}{2\pi L}$

$= \frac{880}{2\pi\times80\times10^{-3}}$

$BW = 1.75\,KHz$

Quality Factor,

$Q = \frac{f_r}{BW}$

$= \frac{2906}{1750}$

$Q = 1.66$

Parallel circuit:

$f_r = \frac{1}{2\pi\sqrt{LC}}$

$f_r = 2.9\,kHz$

Band width,

$BW = \frac{1}{2\pi RC}$

$= \frac{1}{2\times\pi\times880\times0.1\times10^{-6}}$

$BW = 4.82\,KHz$

Quality factor,

$Q = \frac{f_r}{BW} = \frac{2906}{4820} = 0.6$

MARKS SPLIT-UP:

Sl.No	DESCRIPTION	MARKS ALLOTED	MARKS OBTAINED
1.	Experimental setup	20 marks	20
2.	Observation	20 marks	20
3.	Calculation	20 marks	20
4.	Viva	20 marks	18
5.	Record	20 marks	20
6	Total	100 marks.	98

RESULT:

Thus the phenomenon of resonance in RLC circuit was studied and the following were determined using LTspice simulator.

SERIES RLC	PARALLEL RLC
Resonant frequency = 2.9 kHz	Resonant frequency = 2.9 kHz
Bandwidth = 1.75 KHz	Bandwidth = 4.82 KHz
Q factor = 1.66	Q factor = 0.6